

A proof of the conjectured recurrence for OEIS A026005

Adrian Perez Fontelles

Independent researcher

26 August 2026

Abstract

OEIS A026005 carries a conjectured linear recurrence with polynomial coefficients, of order 2. The entry gives no generating function, but it does give the sequence as a single hypergeometric sum. We prove the conjecture without assuming any generating function: creative telescoping produces, from the summand alone, an operator of order 1 that annihilates the sequence, together with a rational certificate that is checked as an exact identity rather than trusted. The conjectured operator is then shown to be a left multiple of it by division in the Ore algebra $\mathbb{Q}(n)[N]$. That last step is what the conjecture needs: the conjectured recurrence has order 2 while the summand only needs order 1, so it is not itself a telescoper and cannot be produced directly.

2020 Mathematics Subject Classification. 33F10, 11B37, 68W30.

1 The sequence and the conjecture

OEIS A026005 is “ $a(n) = T(4^n, n)$, where T = Catalan triangle (A008315)”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 4, 27, 208, 1700, 14364, 123970, 1085760, \dots$$

The entry states, not as a conjecture,

$$a(n) = \text{Sum}\{k=0..n\} ((n+k+1) * \text{binomial}(n+k, k) * \text{binomial}(3^n - k, n-k)) / (2^n + 1). \text{ - Vladimir Kruchinin, Dec 02 2016}$$

so that, writing $F(n, k)$ for the summand,

$$a(n) = \sum_{k=0}^n F(n, k), \quad F(n, k) = \frac{(k+n+1) \binom{-k+3n}{-k+n} \binom{k+n}{k}}{2n+1}. \quad (1)$$

This is the only input the proof takes from the entry, and it was checked against the entry’s own DATA before use.

The entry separately carries the following comment:

$$\textbf{Conjecture: } 9^n * (3^{n+2}) * (3^{n+1}) * a(n) + 12^n * (-55^n * 3 - 59^n * 2 + 65^n * n - 11) * a(n-1) - 32^n * (4^n * n - 5) * (4^n * n - 3) * (2^n * n - 3) * a(n-2) = 0. \text{ - R. J. Mathar, May 22 2013}$$

As of the “Last modified” line on the live entry (2017-05-24, revision 28) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

2 Telescoping the summand

Let N denote the shift in n , $(Nf)(n) = f(n+1)$, and Δ_k the forward difference in k , $(\Delta_k g)(k) = g(k+1) - g(k)$.

Lemma 1. Suppose there are rational functions $\sigma_0, \dots, \sigma_r$ of n , not all zero, and a rational function $R(n, k)$ such that

$$\sum_{i=0}^r \sigma_i(n) F(n+i, k) = \Delta_k [R(n, k) F(n, k)]. \quad (2)$$

Suppose further that $F(n, k) = 0$ for every k outside the range of summation in (1). Then $\sum_i \sigma_i(n) a(n+i) = 0$.

Proof. Because $F(n, k)$ vanishes off the stated range, the sum in (1) may be taken over all $k \in \mathbb{Z}$ without changing it, and the same holds for each $F(n+i, k)$. Summing (2) over all k , the left side becomes $\sum_i \sigma_i(n) a(n+i)$ and the right side telescopes: the partial sums of a forward difference collapse to the boundary values, which are 0 because $R(n, k) F(n, k)$ vanishes for $|k|$ large. Hence the left side is 0. $\square$

The point of Lemma 1 is that (2) is an identity between rational functions once both sides are divided by $F(n, k)$: the quotients $F(n+i, k)/F(n, k)$ and $F(n, k+1)/F(n, k)$ are rational because F is hypergeometric in both variables. So a proposed (σ, R) can be *checked*, exactly, whatever produced it.

For A026005 such a pair exists with $r = 1$:

$$\sigma_0(n) = \frac{-256n^4 - 1152n^3 - 1776n^2 - 1112n - 240}{9n^3 + 30n^2 + 33n + 12}, \quad \sigma_1(n) = 3n + 5$$

and certificate

$$R(n, k) = \frac{9k^5n + 12k^5 - 96k^4n^2 - 192k^4n - 84k^4 + 346k^3n^3 + 856k^3n^2 + 583k^3n + 84k^3 - 712k^2n^4 - 2496k^2n^3 - 2496k^2n^2 - 2496k^2n - 2496k^2 + 36kn^4 + 174kn^3 + 312kn^2 + 246kn + 72k - 3}{9n^3 + 30n^2 + 33n + 12}$$

Dividing (2) by $F(n, k)$ and cancelling gives 0 identically, so Lemma 1 applies and

$$L = \sum_{i=0}^1 \sigma_i(n) N^i \quad (3)$$

annihilates a .

3 From the derived recurrence to the conjectured one

The operator (3) is not the conjectured one. It has order 1, below the conjectured order 2: the conjecture is a consequence of a shorter relation, not the shortest relation itself. The two are connected by division in the Ore algebra $\mathcal{A} = \mathbb{Q}(n)[N]$, where multiplication obeys $Nq(n) = q(n+1)N$.

Lemma 2. If $C = QL$ in $\mathcal{A}$ and $L(a) = 0$, then $C(a) = 0$ at every n where the coefficients of Q are defined.

Proof. $C(a) = (QL)(a) = Q(L(a)) = Q(0) = 0$. The coefficients of Q are rational functions of n , so the step is valid away from their poles. $\square$

Writing the conjectured recurrence as an operator C and dividing by L leaves remainder 0, with quotient

$$Q = \left(\frac{36n^3 + 120n^2 + 132n + 48}{n+2} \right) + (27n^2 + 117n + 126) N.$$

The coefficients of Q are rational functions of n ; their only integer poles are $n \in \{-2\}$, below the range claimed here.

Theorem 3. *The conjectured recurrence*

$$9n(3n+1)(3n+2)a(n) - 12(55n^3 + 59n^2 - 65n + 11)a(n-1) - 32(2n-3)(4n-5)(4n-3)a(n-2) = 0$$

holds for every $n \geq 2$.

Proof. By Lemma 1 the operator (3) annihilates a , and by Lemma 2 applied to $C = QL$ so does the conjectured operator. The integer poles of Q all lie below $n = 2$. $\square$

4 Verification

Every number above is an output of a script written separately from this note, and four independent checks were run.

First, the certificate was verified, not trusted: (2) was divided by $F(n, k)$ and the difference of the two sides cancelled to 0 as a rational function of n and k .

Second, the vanishing of $F(n, k)$ off the range of summation — the hypothesis Lemma 1 needs, and the one that is easy to assume without looking — was checked directly, on concrete integers on both sides of the range.

Third, the factorisation was confirmed by multiplying back: QL was expanded in $\mathcal{A}$ and its coefficients cancelled against those of C to zero.

Fourth, the conjecture itself was evaluated on the published terms, in exact integer arithmetic and without reference to any operator: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 17 values of n from $n = 2$ upward. The formula (1) was likewise checked term by term against the DATA section before being used.

References

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- [4] O. Ore, *Theory of non-commutative polynomials*, Ann. of Math. 34 (1933), 480–508.